Thomas Carr

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EDUCATION

University of Bristol 2026 - present
PhD Economics
London School of Economics 2010 - 2011
MRes Economics *Distinction*
London School of Economics 2009 - 2010
MSc Economics *Distinction*
University of Durham 2004 - 2007
BA(Hons) Economics *First*

RESEARCH FIELDS

Econometric Theory, Causal Inference, Machine Learning

JOB MARKET PAPER

Estimating Local Average Treatment Effects Under Weak Monotonicity

Adstract: We study the estimation of treatment effects with an instrumental variable when there are compliers and defiers. If the two groups exist at distinct covariate values, then the instrumental variable identifies a local average treatment effect, but conventional two-stage least squares does not estimate it. An oracle with knowledge of defier covariate values could correct the two-stage least squares estimator by reordering the instrument wherever defiers are present. We propose a feasible version of this estimator where estimated propensity scores are used to identify defier covariate values. We characterise the asymptotic distribution of the feasible estimator and show conditions under which it is asymptotically equivalent to the oracle estimator. We provide a consistent estimator for the standard error of the feasible estimator, allowing inference. We evaluate the finite sample properties of the feasible estimator by simulation, and find that it outperforms the oracle estimator in terms of mean squared error. We apply the estimator to the data of Card (1993). Compared to conventional two-stage least squares estimates, the point estimates obtained by our estimator differ in sign and are insignificant. These results suggest that the instrument of Card (1993) is weak.

WORK IN PROGRESS

- Instrumental Variables Without Overlap
- Testing Instrument Validity with Covariates (*with Toru Kitagawa*) [Arxiv]
- The Kitagawa (2015) Test for Instrument Validity: Implementation and Extensions (*with Toru Kitagawa*)
- Bayesian Reinforcement Learning for Partially-Observable Markov Decision Processes (*with Kohei Kawaguchi and Toru Kitagawa*)

PRESENTATIONS

IAAE Annual Conference (2022), Siena Workshop on Econometric Theory and Applications (2023), Symposium on Econometric Theory and Applications (2026)

REFERENCES

Sami Stouli	**Weining Wang**	**Toru Kitagawa**
University of Bristol	University of Bristol	Brown University
s.stouli@bristol.ac.uk	weining.wang@bristol.ac.uk	t.kitagawa@brown.edu

TEACHING EXPERIENCE

University of Bristol

MSc Machine Learning for Economics 2026 - 2027
BSc Applied Quantitative Research Methods 2026 - 2027
BSc Probability and Statistics 2026

London School of Economics

BSc Intermediate Macroeconomics 2011 - 2015
Course manager in 2014 - 2015, class teacher prizes in 2012, 2013 and 2014
Summer School Intermediate Macroeconomics 2014
Summer School Introductory Macroeconomics 2013

RESEARCH EXPERIENCE

Brown University 2022 - 2023
Research Associate
University College London 2018 - 2021
Research Assistant
London School of Economics 2011 - 2016
PhD Economics (incomplete)

SKILLS

Computer Languages MATLAB, R, Stata, Python, VBA
Languages English, Japanese (JLPT N1)